

QL SENSOR Series Digital Load Cell Sensor

User Manuals

Standard: GB/T 7551-2008

Our load cell is recommended to be used with the following supporting instruments



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Preface

Thank you for using our QL SENSOR series load cells!

We offer a wide range of load cells and indicators, transmitters and system solutions.

If you are using this type of load cell for the first time, it is important that you read this instruction manual carefully before working with it.

This manual applies to all products of the QL SENSOR range. For special customized products, please read the additional instructions.

To the user

This sensor is best used in conjunction with our digital instruments. For the use of some special instruments and accessories, please consult your supplier or us first.

Do not pull the sensor cable at any time, otherwise the failure will not be covered by the warranty!

Please read the following instructions carefully before use, damage caused by improper operation is not covered by the warranty!

Our company offers a wide range of load cells and indicators, amplifiers, force gauges in many sizes, and has a lot of experience in the field of load cells, and can provide a series of weighing solutions, specific product information can be found by visiting the company's official website,

For business inquiries, please contact: chenming@qlsensor.com

1. Scope of Product Use

The QL SENSOR series load cells produced by our company have excellent load characteristics, temperature characteristics, fully sealed corrosion resistance, strong overload resistance, excellent moisture resistance, high performance explosion-proof, compact structure, wide range of use and other characteristics.

It can be used in various types of electronic belt scales, quantitative coal feeders, hopper scales and other large-scale industrial electronic scales, with various types of high-precision weighing display instruments, automation instruments and large-screen display devices and other types of special electronic weighing equipment.

It can not only be used in the production of agricultural products, warehousing, industrial and mining, trade, processing of materials and other enterprises and institutions, but also widely used in transportation, scientific research and industrial projects supporting projects.

It is easy to install, durable, especially suitable for the working environment is relatively harsh and dust explosion-proof requirements of the occasion.

2. Working principle and circuit diagram

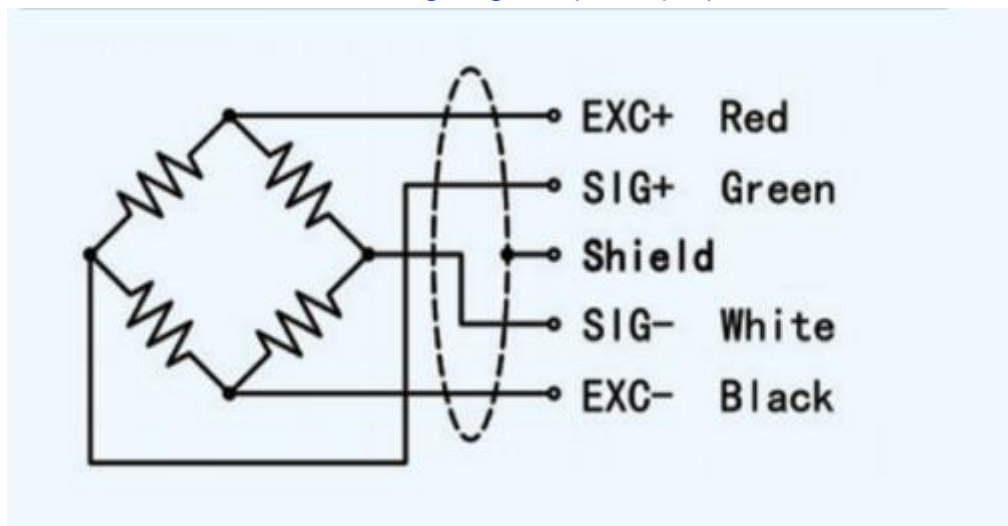
The load cell consists of an elastomer, a resistance strain gage affixed to it and an AD circuit board.

When an external force is applied to the elastomer, the elastomer undergoes elastic deformation. The resistive strain gauges affixed to the elastomer undergo a corresponding deformation.

The output signal of the bridge circuit composed of strain gauges changes, and the output is a millivolt voltage signal, while the analog signal

is sampled by the AD circuit and converted to A/D, and the data processing and corresponding digital compensation measures are carried out to minimize the error of the output. By adjusting the internal input resistance the signal can be changed according to the requirements of the production.

The internal structure of the QL SENSOR series load cell is shown in the following diagram:(example)



3. Main technical indicators

Output sensitivity	1.0-2.0±0.05 mV/V
Non-linearity	±0.03~±0.05 %F · S
Lag error	±0.03 %F · S
Repeatability	±0.03 %F · S
Creep	±0.03 %F · S
Zero temperature coefficient	±0.05 %F · S/10°C
Sensitivity temperature coefficient	±0.05 %F · S/10°C
Input resistance	380±30 Ω
Output resistance	350±5Ω
Working temperature	-20°C~+70°C

Safe overload	150 %F · S
Insulation resistance	$\geq 2000\text{M}\Omega$
Incentive voltage	5V~12V
Protection class	IP66、IP66

4. Installation steps

(1) After opening the package, you should check whether the range of the sensor is consistent with the package, and whether there is any deformation or other bad phenomena caused by the transportation process. (Appearance of slight scratches or color error is a normal phenomenon in the production process, does not affect the normal use of the sensor)

(2) According to the product corresponding to the installation method for fixed, check whether the sensor is installed correctly, according to the correct voltage and range for testing.

(3) Some mini sensors use threaded installation, because the size of the problem may be difficult, please make sure that the use of threads in line with the corresponding specifications, mini sensors in the production process after rigorous testing, if not the use of matching threads there is the possibility of installation difficulties.

5. Precautions and Instructions

Untrained personnel are not permitted to clean, inspect, maintain, repair, or modify the equipment. Always disconnect the equipment from the power supply before cleaning or performing maintenance.

Only qualified personnel are permitted to perform maintenance on this equipment. Be sure to exercise caution when performing power-on

inspections, tests, and adjustments. Failure to observe the above precautions can result in personal injury.

When welding scales, place a grounding clip to prevent welding current from being transmitted to you through the load cell, and be sure to keep the welding equipment as close to the work floor as possible. Never weld within 4 feet (1.2 meters) of the load cell without removing it.

The mounting base requires that the load cell be balanced for installation and that the mounting base for each load cell base be within 0.3 mm of flatness and that the bases be on the same level. The support surfaces connected to the weighing module should be on the same level. If this is not the case, the level should be adjusted with an adjusting spacer. At the same time, use a multimeter to test the output of the sensor to reach within 1/100 error of the actual load to be qualified. When installing loaded parts such as hangers, they must be installed naturally and vertically, and lateral stresses are not allowed.

When the center of gravity of the load is higher than the installation plane, the user should add a limiting device on the load hopper or load plate. It is not permitted to have hard connections around the hopper or load-bearing platform at the installation site to avoid weighing inaccuracies.

The free ends of permanently connected cables should be protected in a manner suitable for explosive atmospheres.

Installation and use on site shall be in accordance with IEC 60079-31.

Installation and use of the load cell on site should be cleaned regularly to avoid dust accumulation.

Faults in the field

---Unstable and jumping display data during unloading or weighing.

After the load cell, junction box and meter are disconnected from the

power supply, re-power them up to see if they automatically return to normal.

----Zero drift

Check whether the cable between the load cell, junction box and meter is broken or damaged, and reconnect the connector if it is connected by a connector.

----No display after loading

Check whether the power supply is stable and whether the power supply voltage is normal.

----The display data is too large when no load.

When using a group of sensors, you need to check whether each sensor is in contact with the loaded object.

----Display data does not return to zero after weighing.

The green and white sensor output wires are disconnected from the junction box or meter; use a multimeter to measure their mV values. The junction box or meter needs to remain powered. From the measured mV value to determine whether it is a sensor failure.

---- Large weighing error

Cut off the power supply, disconnect the input and output lines (four-core cable) of the sensor from the junction box or meter; measure the resistance value of the input (red, black) and output (green, white) with a multimeter; no resistance value can be used to judge the failure of this sensor directly, and judge whether the resistance value is normal against the certificate or manual.

Calculation of sensor accuracy

Calculation of the sensor accuracy is how large kg of the error value of the formula

$$\frac{\text{Accuracy}}{100\%} \times \text{Range} = \pm \text{error value}$$
$$\frac{\text{error value}}{\text{Range}} \times 100\% = \text{Accuracy } (\%)$$

Example 1: sensor accuracy $\pm 0.3\%$ range 20T, 20T will be converted to 20000 kg

$$\frac{0.3}{100} \times 20000 = \pm 60 \text{ (kg)}$$

Example 2: field sensor error 20kg, the maximum range of 5000kg

$$\frac{20}{5000} \times 100\% = 0.4\%$$

The above method can easily calculate the accuracy of the actual test in the field.

Load cells using strain gauges are very sensitive and can detect very small changes in weight. The trick is to ensure that they only respond to the weight you want to measure and not to other forces. To get an accurate weight reading, you must carefully verify how and where the weight is applied to the load cell. In theory, the load cell should be installed so that the load is applied vertically over the entire weight range (see Figure 5-1).

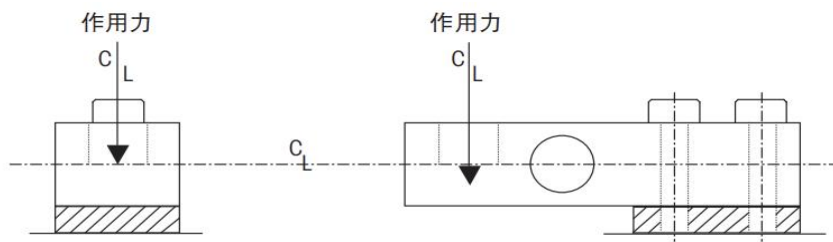


图 5-1：理想负载（作用力完全垂直）

For ideal weighing results, the weigh container and load cell supports need to be level and parallel, and must be incredibly strong.

If the platform and its structural supports are carefully designed and installed, the scale will achieve the desired load application.

If they are not installed correctly, a variety of forces can affect their accuracy.

6. Use of the working environment

Protect against general corrosion, radio frequency interference, lightning strikes and static electricity. It can be measured normally when the working temperature of the environment is $-20^{\circ}\text{C}\sim+70^{\circ}\text{C}$.

7. Safety Precautions

1) The input voltage is $5\text{V}\sim 12\text{V}$, if it is lower than 5V , it may lead to abnormal output, if it is higher than 12V , it may damage the sensor.

2) Strong impacts at the loading end may cause permanent damage to the sensor.

3) Mechanical collision and friction may reduce the measurement accuracy and service life of the sensor.

4) Sensors that have been disassembled by the user may not be covered by a good warranty.

8. Maintenance matters

In order to facilitate maintenance, if you suspect that the sensor is damaged during use, please contact us in time so that we can provide you with better service.

After-sales e-mail: chenming@qlsensor.com

9. After-sales service

The following conditions do not enjoy the after-sales service of the product:

The product structure is incomplete, any reverse engineering or private disassembly, modification behavior, or the existence of obvious impact on product safety, reliability and function, performance of the appearance of damage. Storage and use conditions do not meet the specified requirements, resulting in failure or misalignment

Attachment: packing list

- 1 Load Cell Sensor
- 1 Certificate of Conformity
- 1 Instruction manual

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